

Nikhil Prabhu

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EDUCATION

University of California, San Diego (UCSD)
B.S. in Computer Science | Cumulative GPA: 3.67

San Diego, CA
Expected Graduation: June 2028

SKILLS

Languages: Python, C++, TypeScript, JavaScript

Data + Embodied AI: egocentric/robot data pipelines, MediaPipe pose, depth, behavioral cloning, PPO/MAPPO, VLM verification

Robotics: ROS 2, SLAM, visual odometry, MuJoCo, embedded/hardware integration, GPS-denied navigation

Backend + Infra: FastAPI, Node.js, WebSockets, Supabase, Docker, Git, Linux, Google Cloud Run, real-time streaming pipelines

EXPERIENCE

Undergraduate Research Assistant — AI for Circuit Invention, UCSD

San Diego, CA

Research Assistant (with Prof. Prasad Gudem & Samvid AI)

March 2026 – Present

- Built data infrastructure connecting LLM APIs to Cadence simulation tools, automating a generate-simulate-evaluate loop and parsing simulator output into structured benchmarks for downstream training.
- Curated and labeled multimodal design samples (netlists, simulation logs, performance metrics), turning raw tool output into clean supervision signals for model fine-tuning.

Desktop Agent Developer & Founder — UCSD Agent Development (Clue2)

San Diego, CA

Mentored by Prof. Ramamohan Paturi

Jan 2025 – Present

- Built a cross-platform Electron agent with real-time meeting transcription, WebSocket streaming, buffering, and fault-tolerant sessions over live sensor/audio streams.
- Shipped an LLM workflow engine across Gmail, Calendar, Docs, Sheets, Slack, Notion, and 7+ services; scaled beta to 40+ users with YC-startup adoption and session metadata for training traces.

Lead Researcher — Brain-Computer Interface (Stanford Lee-Messer Lab)

Stanford, CA

Stanford University School of Medicine

Oct 2023 – Dec 2024

- Collected, labeled, and benchmarked multimodal EEG datasets on a P300 paradigm; piloted with 12 users and shipped a React Native frontend turning raw neural signals into labeled, evaluation-ready control output.

ROBOTICS & DATA PROJECTS

SODIUM — Embodied Robot

1st Overall, DiamondHacks 2026

April 2026

- Built a ROS 2 robot stack with wake-word detection, NL task routing, medication reminders, and crisis detection, integrating on-device perception with a cloud caregiver dashboard (Svelte 5 / Bun).

DroneOS

1st Overall, Bow Capital × DS3 × SIC Hackathon (UCSD)

2026

- Built a GPS-denied platform with SLAM, visual odometry, Gaussian Splatting maps, and YOLOX + Norfair perception; trained swarm policies (MAPPO, behavioral cloning) over large rollout datasets without comms or GPS.

Battle Angel — Simulation Data Generation

Google Gemini 3.5 Hackathon

2026

- Curated rollout data across eight MuJoCo disaster environments via vectorized PPO + FastAPI with GIF replay and Gemini-generated scene layouts for a Unitree G1 humanoid.

Motion — Crowdsourced Robot Training Data Marketplace

Stanford × DeepMind Hackathon

2026

- Built an egocentric pipeline where contributors record industrial manipulation clips via phone with live MediaPipe skeleton overlay and instant pose-confidence scoring.
- Designed a Gemma 4 + Kimi K2 VLM verification stack on Google Cloud Run and exported LeRobot / Open X-Embodiment bundles — turning human demonstration video into clean, labeled embodied-AI training data.

PUBLICATIONS

- **Real-Time EMG Control for Assistive Devices** — IEEE ICHCI 2024.
- **Co-Heritability of Antibiotic Resistance in TB Evolution** — Bioinformatics & Biomedical Technology, 2025.